

TrES-5b

R-0897

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250813 · created 2026-07-29T03:08:25+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460900.8879 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.053
Transit depth (Rp/Rs)^2	3.34 +/- 1.92 [%]
Orbital Inclination (inc)	82.3 +/- 2.85
Airmass coefficient 1 (a1)	1.2 +/- 0.19
Airmass coefficient 2 (a2)	-0.13 +/- 0.14
Scatter in the residuals of the lightcurve fit is	16.1 %
Best Comparison Star	#1 - [317, 406]
Optimal Aperture	1.77
Optimal Annulus	5.37
Transit Duration (day)	0.058 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.053 vs 0.142 (deviation 0.77σ)
Duration fit vs published	0.058 d vs 0.0758 d (deviation 0.66σ)
Mid-time O–C	3.9 min
Depth SNR	3.5
Residual scatter	16.1 %

Light curve

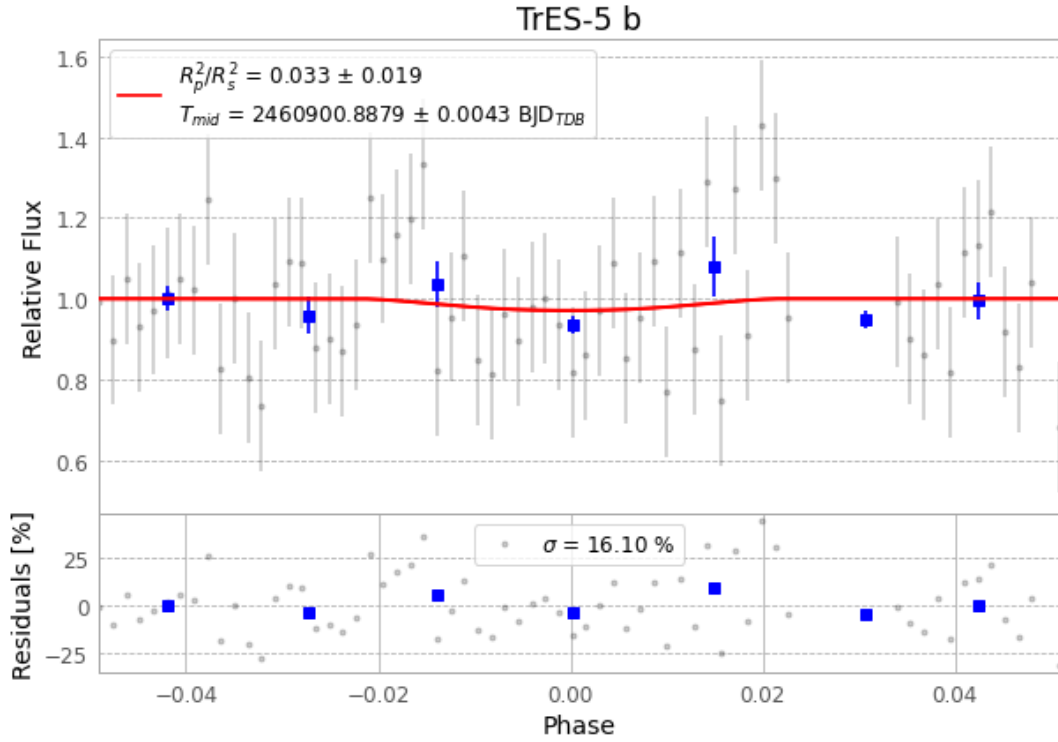


Figure 1 — FinalLightCurve_TrES-5 b_2025-08-13.png (SHA-256 15db9dd2ae89e637...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 268], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a4d42b143a799cf0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5250813073016.FITS → TRES-5250813110315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250813.log (4170eb8e400b...), FinalLightCurve_TrES-5 b_2025-08-13.png (15db9dd2ae89...), FinalLightCurve_TrES-5 b_2025-08-13.csv (c1183086dab8...)

Compute resources

Wall time 190.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 03:08Z.